Retail POS Transaction Analysis

Summary Findings Report — Multi-Store Retail Performance, 2024–2025

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Date: August 12, 2026
Data source: POS transactions, 30 stores (MySQL)

Executive Summary

This report summarizes findings from an analysis of 970,838 POS transaction line items (576,242 orders) across our 30-store network for 2024 and 2025. The goal was to understand overall business performance and identify where the business has the biggest opportunities for improvement. The analysis covers revenue and profitability, store and category performance, product mix, customer behavior, returns, and payment methods.

$46.60M	$25.78M	55.34%	~$79.50
Net Sales (2024–25)	Gross Profit	Gross Margin	Avg Order Value

The headline takeaway: the business grew from 2024 to 2025, but that growth came almost entirely from more transactions rather than customers spending more per visit. The single biggest opportunity uncovered is that we cannot identify the customer behind roughly two-thirds of our sales, which limits our ability to run any kind of targeted retention or loyalty program.

1. Overall Business Performance

Finding: Net sales totaled $46.60M and gross profit totaled $25.78M across 2024–2025, for an overall gross margin of 55.34%. The business processed 576,242 payment transactions and sold 1,173,116 net units (after refunds).

Recommendation: These figures give us a reliable baseline. Future reporting should track net sales, gross margin, and transaction count together, since margin trends can be masked by revenue growth alone.

Finding: The average order contains about 1.65 line items — most orders (327,448) contain a single item, with a fast drop-off for larger baskets.

Recommendation: There may be a meaningful upside in basket-building tactics (bundling, cross-sell prompts at checkout, "complete the look" merchandising) since the average customer is currently buying close to one item per visit.

2. Yearly Performance

Year	Net Sales	Transactions	Gross Profit	AOV	Gross Margin
2024	$22,675,326.92	284,791	$12,544,958.43	$79.62	55.32%

Year	Net Sales	Transactions	Gross Profit	AOV	Gross Margin
2025	$23,919,846.35	300,900	$13,239,097.91	$79.49	55.35%

Finding: Net sales grew 5.5% year over year (from $22.68M to $23.92M), driven by a 5.7% increase in transactions (284,791 to 300,900). Average order value stayed essentially flat, moving from $79.62 to $79.49.

Recommendation: Our growth strategy last year worked from a traffic/transaction-volume standpoint. Going forward, pairing that with initiatives to lift average order value (e.g. upsell training, bundle pricing) could meaningfully compound our growth rate rather than relying on transaction volume alone.

3. Store Performance

Finding: Growth varies widely across our 30 stores. The strongest performers were Store 23 – Baltimore (+9.82% sales, +3.28% AOV), Store 20 – New York (+9.09% sales, +2.58% AOV), and Store 07 – Seattle (+7.12% sales, +1.92% AOV). Store 13 – Miami stood out in the other direction: sales declined 0.10% even though transactions grew 3.23%, because AOV fell 3.23%.

Recommendation: Baltimore, New York, and Seattle are worth a closer look as potential playbooks for the rest of the network — their growth is coming from customers spending more per visit, not just more visits. Miami's declining AOV despite more foot traffic suggests a pricing, promotion, or assortment issue worth investigating directly with that store's management.

4. Category & Product Performance

Finding: Apparel is our largest category by revenue ($17.91M), followed by Footwear ($14.32M). However, Beauty has the highest gross margin at 68.22%, while Footwear has both the lowest margin (48.68%) and the highest cost percentage (51.32% of net sales).

Recommendation: Apparel and Footwear should remain shelf-space priorities given their revenue scale, but Beauty's strong margin profile makes it a good candidate for expanded assortment, more prominent placement, or increased promotional investment — it is quietly one of our most efficient categories.

Finding: Our top 5 products account for 43.86% of net sales and 39.65% of gross profit; the top 10 account for roughly 69% of sales and 65% of profit. Running Shoes alone represents 10.46% of net sales and 8.92% of gross profit.

Recommendation: A small number of SKUs carry an outsized share of the business. We should prioritize inventory availability and supplier terms for these top items specifically — a stockout on Running Shoes, for example, has a much bigger revenue impact than a stockout on a typical item.

5. Customer Analysis

Finding: Only 19,999 customers are identifiable in our data, and they account for just 35.33% of net sales — the remaining 64.67% of revenue comes from transactions with no linked customer.

Recommendation: This is the largest opportunity surfaced in this analysis. We are currently unable to run targeted retention, win-back, or loyalty campaigns against roughly two-thirds of our revenue because we don't know who generated it. Capturing customer identification more consistently at checkout (loyalty sign-up, phone or email capture, app usage) should be a priority initiative.

Finding: Among the customers we can identify, repeat purchasing is overwhelmingly the norm — 19,993 are repeat customers versus only 6 one-time customers. Identified customers also have a slightly higher AOV ($81.62) than unidentified transactions ($80.45), and average 10.08 transactions over the period.

Recommendation: This tells us that once we do capture a customer's identity, they tend to be loyal and spend consistently. That strengthens the case for investing in identification: it is not a shot in the dark — the customers we can reach are already engaged.

Finding: Customer value scales strongly with purchase frequency. Very High Frequency customers average $1,385.01 in sales (17.03 transactions on average), compared to $1,015.78 for High Frequency customers. High-frequency customers concentrate their spend in Apparel ($2.93M) and Footwear ($2.35M).

Recommendation: Loyalty and re-engagement programs are well worth the investment — frequency clearly compounds into customer value. Apparel and Footwear are the natural categories to anchor loyalty offers and retention campaigns around, since that's where our most frequent customers already shop.

6. Returns & Refunds

Finding: Return rates are low and consistent across categories, ranging narrowly from about 0.78% to 0.82% by quantity. Apparel has the lowest return rate; Home Goods and Footwear have the highest, though the gap is minor.

Recommendation: Returns are not currently a meaningful profitability concern anywhere in the business. This area can be monitored on a routine cadence rather than treated as an active priority.

7. Payment Method Analysis

Finding: Card is by far the dominant payment method, representing 70.93% of net sales ($33.05M across 409,143 transactions). Cash makes up 24.04%, and other methods make up the remaining 5.03%. Average transaction value is nearly identical across all payment methods ($80.78–$81.40).

Recommendation: Payment method does not meaningfully influence how much customers spend, so it isn't a lever for growing AOV. Given card's dominant share, continued investment in fast and reliable card processing at checkout is worthwhile to avoid friction on the majority of our transaction volume.

Summary of Recommendations

- **Prioritize customer identification at checkout** — the biggest opportunity in this analysis; roughly two-thirds of revenue is currently untraceable to a customer.
- **Study and replicate what's working in Baltimore, New York, and Seattle** — these stores are growing AOV, not just traffic.
- **Investigate Miami's declining AOV** directly with store management.
- **Protect inventory availability for top-performing SKUs**, especially Running Shoes and the rest of the top 10 products.
- **Expand investment in Beauty** given its strong margin profile relative to Apparel and Footwear.
- **Build loyalty offers around Apparel and Footwear**, where high-frequency customers already concentrate spend.
- **Explore basket-building tactics** (bundling, cross-sell) given the average order is close to a single item.

Methodology: Raw POS data was cleaned and validated in Python (pandas), loaded into a MySQL database, and analyzed using SQL (aggregations, CTEs, and window functions). Findings were visualized in a supporting Power BI dashboard. Full query documentation is available in the accompanying project repository.